Preprocessing

Medical Report
The cardiac, mediastinal and hilar contours are normal. Pulmonary vasculature is normal. Lungs are clear.
...

CheXbert

RadGraph

Label
Enlarged Cardiomediastinum_F
Pneumothorax_F
Pleural Effusion_F
...

Triple
(contours, modify, cardiac)
(contours, modify, hilar)
(normal, located_at, cardiac)
(normal, located_at, mediastinal)
...

LLM-SA

Semantic Clustering

Enlarged Cardiomediastinum_F
(contours, modify, cardiac)
(normal, located_at, cardiac)
.....

Pneumothorax_F
(clear, located_at, lungs)
.....

Pleural Effusion_F
(effusion, located_at, pleural)
.....

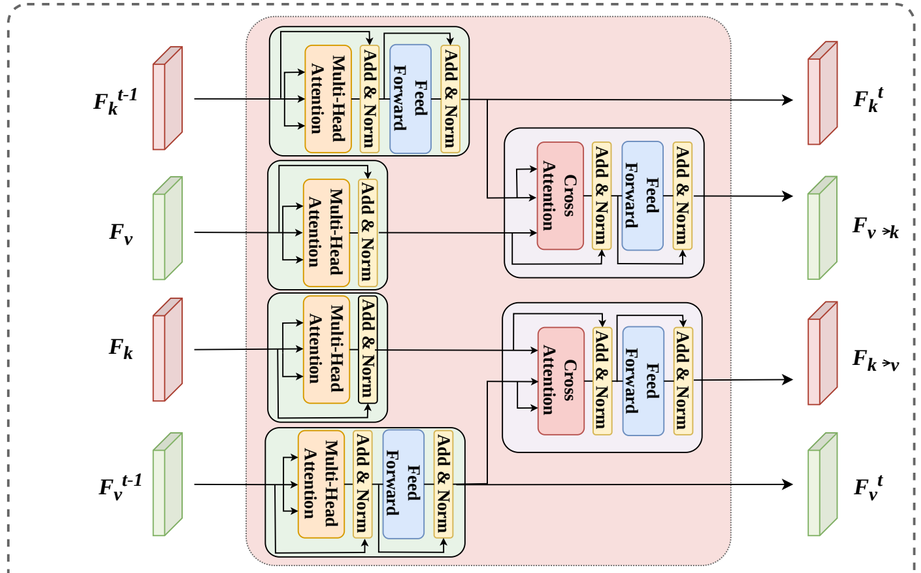
Other
.....

...

Integration

KG

Knowledge Graph Construction (KGC)



Causal Interaction Fusion (CIF)

Model Training

X-Ray Image



Visual Causal Encoder

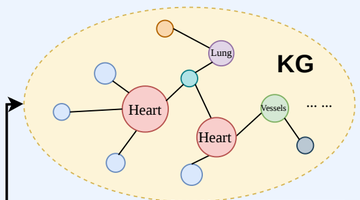
Retrieval

Classifier

Top-K Selection

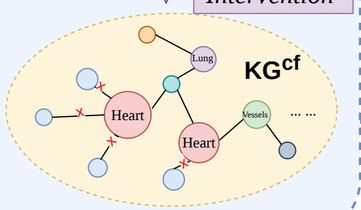
KG

SCR



Causal

Intervention



Medical Report

Text Encoder

Ft

F_v

$F_{v \to k}$

CIF

F_k

$F_{k \to v}$

KG Encoder

F_k^{cf}

$F_{v \to k}^{cf}$

CIF

$F_{k \to v}^{cf}$

Decoder

Masked Multi-Head Attention

Add & Norm

Multi-Head Attention | Multi-Head Attention

Add & Norm | Add & Norm

Causal Gating Unit

Feed Forward

Add & Norm

Linear

Softmax

Factual Report Generation

Counterfactual Report Generation

Model Inference



Visual Causal Encoder

Retrieval

KG

SCR

KG Encoder

F_k F_k^{cf} F_v

CIF CIF

Decoder

Report Generation